

Preprocesamiento

2025-10-13

Datos y eleccion de herramientas:

Para la etapa inicial del proyecto se optó por trabajar con QGIS y la consola de R con el objetivo de realizar un preprocesamiento previo al Analisis de los datos espaciales.

La elección de estas herramientas se basó en que ya se contaba con desarrollos y rutinas previas en este entorno, lo que permitió optimizar tiempos y asegurar la coherencia espacial de la información.

De esta forma, se priorizó el uso de QGIS para la inspección visual, validación y limpieza de geometrías, y R (con librerías como sf y dplyr, h3jsr, automap y gstat) para la generación de grillas hexagonales, cálculos de área y la estructuración de los atributos.

Esta instancia permitió dejar los datos listos y estandarizados para la etapa 2 del proyecto

Grilla H3 formato Poligono: a partir del limite del lote se genero los hexagonos con resolucion 12

```
setwd("C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh")
library(sf)
library(dplyr)
library(h3jsr)
```

```
Shape <- sf::st_read("la_magdalena_L4.geojson")
```

```
## Reading layer 'la_magdalena_L4' from data source
##   'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\la_magdalena_L4.geojson'
##   using driver 'GeoJSON'
## Simple feature collection with 1 feature and 11 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XYZ
## Bounding box:   xmin: -63.80942 ymin: -33.99367 xmax: -63.78875 ymax: -33.98186
## z_range:        zmin: 0 zmax: 0
## Geodetic CRS:   WGS 84
```

```
SRC <- st_crs(Shape)
Limite_SinZ <- st_zm(Shape)
Limite_SinZ <- st_make_valid(Limite_SinZ)
File_rep <- st_transform(Limite_SinZ, crs = 4326)

Resolucion_H3 <- 12          # Resolución
File_rep_hex <- polygon_to_cells(File_rep, res = Resolucion_H3)
```

```

Hex_H3_sf <- h3jsr::cell_to_polygon(input = File_rep_hex, simple = FALSE)

# Variables de ejemplo:
Cliente      <- "Garruchos agropecuaria"
Season       <- "2019-2020"
Region_CREA  <- "Centro"
Farm         <- "La Magdalena"
Field        <- "L4"
Cultivo      <- "Soja"
Hibrido      <- "SojaX"

Hex_H3_sf <- Hex_H3_sf %>%
  mutate(
    Empresa      = Cliente,
    Campania     = Season,
    Region       = Region_CREA,
    Campo        = Farm,
    Lote         = Field,
    Actividad    = Cultivo,
    Material     = Hibrido,
    Hexagono     = row_number(),
    Id_Hexagono  = h3_address
  )

# Calcular áreas
Hex_H3_sf <- Hex_H3_sf %>%
  mutate(
    Area_lote     = round(as.numeric(st_area(Limite_SinZ) / 10000), 3),
    Area_hexagono = round(as.numeric(st_area(geometry) / 10000), 5)
  )

centroids <- st_centroid(Hex_H3_sf)
Hex_H3_sf <- Hex_H3_sf %>%
  mutate(
    Long = st_coordinates(centroids)[,1],
    Lat  = st_coordinates(centroids)[,2]
  )

orden_columnas <- c("Empresa", "Campania", "Region", "Campo", "Lote",
  "Actividad", "Material", "Area_lote", "Hexagono",
  "Id_Hexagono", "Area_hexagono", "Lat", "Long")

cols_existentes <- intersect(orden_columnas, names(Hex_H3_sf))
Hex_H3_sf <- Hex_H3_sf %>% select(all_of(cols_existentes), geometry)

# Reproyectar
Grid <- st_transform(Hex_H3_sf, crs = SRC)

print(head(Grid))

```

```

## Simple feature collection with 6 features and 13 fields
## Geometry type: POLYGON
## Dimension:      XY

```

```
## Bounding box: xmin: -63.80615 ymin: -33.99274 xmax: -63.79044 ymax: -33.98417
## Geodetic CRS: WGS 84
##      Empresa  Campania Region      Campo Lote Actividad Material
## 1 Garruchos agropecuaria 2019-2020 Centro La Magdalena L4      Soja      SojaX
## 2 Garruchos agropecuaria 2019-2020 Centro La Magdalena L4      Soja      SojaX
## 3 Garruchos agropecuaria 2019-2020 Centro La Magdalena L4      Soja      SojaX
## 4 Garruchos agropecuaria 2019-2020 Centro La Magdalena L4      Soja      SojaX
## 5 Garruchos agropecuaria 2019-2020 Centro La Magdalena L4      Soja      SojaX
## 6 Garruchos agropecuaria 2019-2020 Centro La Magdalena L4      Soja      SojaX
##   Area_lote Hexagono      Id_Hexagono Area_hexagono      Lat      Long
## 1    243.618        1 8cc34d60575ebff      0.02385 -33.98731 -63.80324
## 2    243.618        2 8cc34d601ce8dff      0.02385 -33.98425 -63.79053
## 3    243.618        3 8cc34d600af5bff      0.02384 -33.99266 -63.79404
## 4    243.618        4 8cc34d605393dff      0.02385 -33.98680 -63.80606
## 5    243.618        5 8cc34d600a76dff      0.02384 -33.99231 -63.79645
## 6    243.618        6 8cc34d6057b17ff      0.02385 -33.99179 -63.80199
##      geometry
## 1 POLYGON ((-63.80326 -33.987...
## 2 POLYGON ((-63.79055 -33.984...
## 3 POLYGON ((-63.79406 -33.992...
## 4 POLYGON ((-63.80607 -33.986...
## 5 POLYGON ((-63.79647 -33.992...
## 6 POLYGON ((-63.80201 -33.991...
```

Campañas de Soja:

Se separan las 3 campañas por separado: 2019, 2021 y 2023

```
setwd("C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh")
library(sf)
library(dplyr)
```

```
gdf <- st_read("soy_performance_2019_2021_2023.geojson")
```

```
## Reading layer 'soy_performance_2019_2021_2023' from data source
##   'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\soy_performance_2019_2021_2023.geojson'
##   using driver 'GeoJSON'
## Simple feature collection with 30684 features and 3 fields
## Geometry type: POINT
## Dimension:      XY
## Bounding box:  xmin: -63.80942 ymin: -33.99365 xmax: -63.78875 ymax: -33.98182
## Geodetic CRS:  WGS 84
```

```
print(names(gdf))
```

```
## [1] "h3_res12"      "campaign_year" "performance_pct" "geometry"
```

```
# Renombrar columnas con sufijo
add_suffix <- function(df, year) {
  df %>%
    rename_with(~paste0(., "_", year), .cols = !c("geometry"))
}
```

```

}

# Filtrar por campaña
gdf_2019 <- gdf %>% filter(campaign_year == 2019) %>% add_suffix(2019)
gdf_2021 <- gdf %>% filter(campaign_year == 2021) %>% add_suffix(2021)
gdf_2023 <- gdf %>% filter(campaign_year == 2023) %>% add_suffix(2023)

# Exportar
st_write(gdf_2019, "soy_performance_2019.geojson", delete_dsn = TRUE)

## Deleting source 'soy_performance_2019.geojson' using driver 'GeoJSON'
## Writing layer 'soy_performance_2019' to data source
## 'soy_performance_2019.geojson' using driver 'GeoJSON'
## Writing 10241 features with 3 fields and geometry type Point.

st_write(gdf_2021, "soy_performance_2021.geojson", delete_dsn = TRUE)

## Deleting source 'soy_performance_2021.geojson' using driver 'GeoJSON'
## Writing layer 'soy_performance_2021' to data source
## 'soy_performance_2021.geojson' using driver 'GeoJSON'
## Writing 10249 features with 3 fields and geometry type Point.

st_write(gdf_2023, "soy_performance_2023.geojson", delete_dsn = TRUE)

## Deleting source 'soy_performance_2023.geojson' using driver 'GeoJSON'
## Writing layer 'soy_performance_2023' to data source
## 'soy_performance_2023.geojson' using driver 'GeoJSON'
## Writing 10194 features with 3 fields and geometry type Point.

```

Join espacial:

Considerando que los datos de Performance de soja de las diferentes campañas estan vinculados al centroide de cada hexagono de la Grilla H3, se realizo un Join espacial de manera de juntar esa informacion en una sola capa y facilitar posteriores analisis.

```

# Directorio
setwd("C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh")
library(sf)
library(dplyr)

#grid <- st_read("Grid.geojson") # Debe tener Id_Hexagono
grid <- Grid

soy_2019 <- st_read("soy_performance_2019.geojson")

## Reading layer 'soy_performance_2019' from data source
## 'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\soy_performance_2019.g
## using driver 'GeoJSON'
## Simple feature collection with 10241 features and 3 fields

```

```
## Geometry type: POINT
## Dimension:      XY
## Bounding box:   xmin: -63.80942 ymin: -33.99357 xmax: -63.78875 ymax: -33.98182
## Geodetic CRS:   WGS 84
```

```
soy_2021 <- st_read("soy_performance_2021.geojson")
```

```
## Reading layer 'soy_performance_2021' from data source
##   'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\soy_performance_2021.g
##   using driver 'GeoJSON'
## Simple feature collection with 10249 features and 3 fields
## Geometry type: POINT
## Dimension:      XY
## Bounding box:   xmin: -63.80942 ymin: -33.99365 xmax: -63.78876 ymax: -33.98182
## Geodetic CRS:   WGS 84
```

```
soy_2023 <- st_read("soy_performance_2023.geojson")
```

```
## Reading layer 'soy_performance_2023' from data source
##   'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\soy_performance_2023.g
##   using driver 'GeoJSON'
## Simple feature collection with 10194 features and 3 fields
## Geometry type: POINT
## Dimension:      XY
## Bounding box:   xmin: -63.80942 ymin: -33.99357 xmax: -63.78876 ymax: -33.98191
## Geodetic CRS:   WGS 84
```

```
# Función
agregar_al_grid <- function(grid_sf, soy_sf) {
  # Join espacial: asignar hex a cada punto
  joined <- st_join(soy_sf, grid_sf[, "Id_Hexagono"], join = st_within)

  agregados <- joined %>%
    st_drop_geometry() %>%
    group_by(Id_Hexagono) %>%
    summarise(across(everything(), ~ mean(.x, na.rm = TRUE), .names = "{.col}")) %>%
    ungroup()

  grid_sf <- left_join(grid_sf, agregados, by = "Id_Hexagono")
  return(grid_sf)
}
```

```
# Aplicar a cada capa
grid_2019 <- agregar_al_grid(grid, soy_2019)
grid_2021 <- agregar_al_grid(grid_2019, soy_2021) # ya acumula 2019
grid_2023 <- agregar_al_grid(grid_2021, soy_2023) # acumula 2019+2021+2023

st_write(grid_2023, "soy_performance_completo.geojson", delete_dsn = TRUE)
```

```
## Deleting source 'soy_performance_completo.geojson' using driver 'GeoJSON'
## Writing layer 'soy_performance_completo' to data source
##   'soy_performance_completo.geojson' using driver 'GeoJSON'
## Writing 10214 features with 22 fields and geometry type Polygon.
```

Exploración y limpieza de datos (Outliers):

Para la capa veris_data, se realizó una etapa de exploración y limpieza de datos con el objetivo de identificar posibles valores atípicos (outliers) que pudieran afectar los análisis posteriores.

Se elaboraron gráficos de histograma y boxplot a fin de visualizar la distribución de las variables y detectar comportamientos anómalos. A partir de esta inspección, se aplicaron filtros simples de outliers, eliminando aquellos valores que se encontraban fuera del intervalo de confianza definido.

```
# Histograma y boxplot de las tres variables:  
# Se aplican para las 3 variables
```

```
setwd("C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh")  
veris <- st_read("veris_data.geojson")
```

```
## Reading layer 'veris_data' from data source  
##   'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\veris_data.geojson'  
##   using driver 'GeoJSON'  
## Simple feature collection with 15383 features and 4 fields  
## Geometry type: POINT  
## Dimension:      XY  
## Bounding box:   xmin: -63.80939 ymin: -33.99356 xmax: -63.7888 ymax: -33.98196  
## Geodetic CRS:   WGS 84
```

```
Layer <- st_read("veris_data.geojson")
```

```
## Reading layer 'veris_data' from data source  
##   'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\veris_data.geojson'  
##   using driver 'GeoJSON'  
## Simple feature collection with 15383 features and 4 fields  
## Geometry type: POINT  
## Dimension:      XY  
## Bounding box:   xmin: -63.80939 ymin: -33.99356 xmax: -63.7888 ymax: -33.98196  
## Geodetic CRS:   WGS 84
```

```
Parameter <- "ec_surface"  
if (!(Parameter %in% names(Layer))) stop(paste("La columna", Parameter, "no existe."))
```

```
# Convertir a numérico y eliminar NA  
valores <- as.numeric(Layer[[Parameter]])  
valores <- na.omit(valores)
```

```
# Layout: 2 filas, 1 columna  
layout(matrix(c(1,2), nrow = 2, byrow = TRUE), heights = c(2,1))  
par(mar = c(4,4,2,1)) # margenes
```

```
# Histograma  
hist(  
  valores,  
  main = paste("Histograma -", Parameter),  
  xlab = "Linea solida = media // linea punteada = mediana",  
  ylab = "Frecuencia",  
  breaks = 25,
```

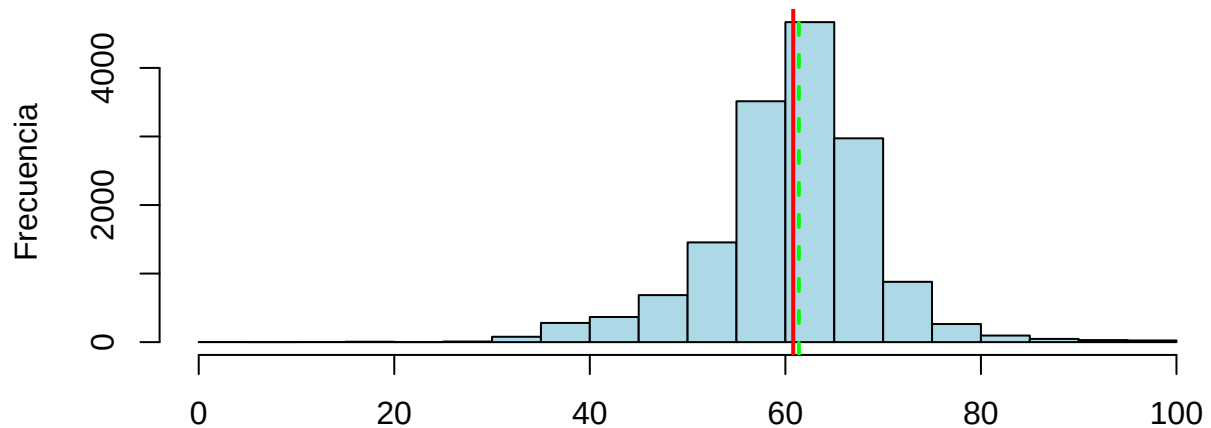
```

col = "lightblue",
cex.lab = 1,
cex.axis = 1,
cex.main = 1.2
)
abline(v = mean(valores), col = "red", lwd = 2)
abline(v = median(valores), col = "green", lty = 2, lwd = 2)

# Boxplot
boxplot(
  valores,
  main = paste("Boxplot -", Parameter),
  horizontal = TRUE,
  xlab = Parameter,
  col = "lightyellow",
  cex.lab = 1,
  cex.axis = 1,
  cex.main = 1.2
)
abline(v = mean(valores), col = "red", lwd = 2)
abline(v = median(valores), col = "green", lty = 2, lwd = 2)

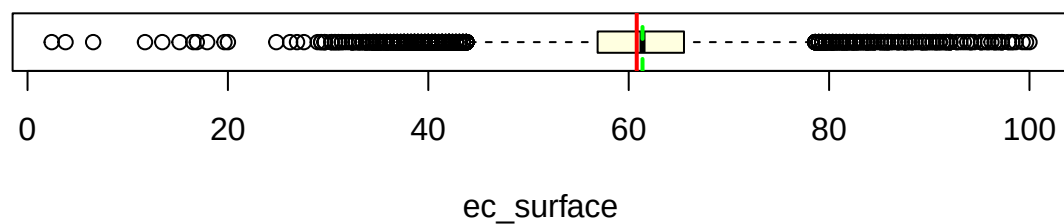
```

Histograma – ec_surface



Linea solida = media // linea punteada = mediana

Boxplot – ec_surface



```

# Outliers:
# Se aplican para las 3 variables.
setwd("C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh")

```

```
veris <- st_read("veris_data.geojson")
```

```
## Reading layer 'veris_data' from data source
## 'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\veris_data.geojson'
## using driver 'GeoJSON'
## Simple feature collection with 15383 features and 4 fields
## Geometry type: POINT
## Dimension: XY
## Bounding box: xmin: -63.80939 ymin: -33.99356 xmax: -63.7888 ymax: -33.98196
## Geodetic CRS: WGS 84
```

```
# Función para eliminar outliers:
sinoutli <- function(mydat, times_sd = 3, yielcol = "VRYIELDMAS") {
  if (!(yelcol %in% colnames(mydat))) {
    stop(paste("La columna", yielcol, "no existe en los datos."))
  }

  n_initial <- nrow(mydat)
  mydat <- mydat[mydat[[yelcol]] != 0, ]
  n_after_zero <- nrow(mydat)
  cat("Se eliminaron", n_initial - n_after_zero, "filas con valor cero.\n")

  Mean <- mean(mydat[[yelcol]], na.rm = TRUE)
  stde <- sd(mydat[[yelcol]], na.rm = TRUE)

  Lower <- Mean - times_sd * stde
  Upper <- Mean + times_sd * stde

  mydat[[yelcol]][mydat[[yelcol]] < Lower | mydat[[yelcol]] > Upper] <- NA

  sinout <- mydat[!is.na(mydat[[yelcol]]), ]
  n_final <- nrow(sinout)
  cat("Se eliminaron", n_after_zero - n_final, "outliers. Quedaron", n_final, "puntos.\n")

  return(sinout)
}

# Aplicar:
veris_sin_outliers <- sinoutli(veris, times_sd = 2.5, yielcol = "ec_surface")
```

```
## Se eliminaron 0 filas con valor cero.
## Se eliminaron 503 outliers. Quedaron 14880 puntos.
```

```
# Exportar:
st_write(veris_sin_outliers, "veris_sin_outliers.geojson", delete_dsn = TRUE)
```

Interpolacion kriging:

La capa de hexágonos o los puntos “soy_performance_2019_2021_2023” no representa adecuadamente la distribución espacial de los datos de la capa “veris_data”. La distribución de los puntos “veris_data” están

dispersos y dejan grandes áreas sin información dentro de cada hexágono. Por eso, necesito aplicar una interpolación. Esto me permitirá crear una superficie continua que represente mejor la variación de los datos en toda el lote y dentro de cada hexágono. Así, se podrá analizar las variables de forma más precisa y obtener resultados más confiables.

```
setwd("C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh")

library(sf)
library(gstat)
library(sp)
library(automap)

kriging_ec_surface <- function(file_path) {
  cat("Procesando:", file_path, "\n")

  data <- st_read(file_path, quiet = TRUE)
  data <- st_make_valid(data)

  # Verificar columna
  if (!"ec_surface" %in% names(data)) stop("La columna 'ec_surface' no existe.")
  data$ec_surface <- as.numeric(data$ec_surface)
  data <- data[!is.na(data$ec_surface), ]

  # Reproyectar a EPSG:32720
  data <- st_transform(data, 32720)

  data_sp <- as(data, "Spatial")

  # Ajuste del variograma automático
  vgm_model <- autofitVariogram(ec_surface ~ 1, data_sp, cutoff = 250)

  # Crear grilla desde bounding box
  bbox <- st_bbox(data)
  grid_points <- st_make_grid(
    st_as_sfc(bbox),
    cellsize = 5, # metros
    what = "centers"
  )

  if (length(grid_points) < 10) stop("la malla aún tiene pocos puntos (", length(grid_points), ").")

  # Interpolacion Kriging
  kriging_result <- krige(
    formula = ec_surface ~ 1,
    locations = data_sp,
    newdata = grid_points,
    model = vgm_model$var_model,
    nmax = 25,
    nmin = 7,
    maxdist = 40
  )

  # Convertir resultado a sf
  kriging_sf <- st_as_sf(kriging_result)
```

```
kriging_sf <- st_set_crs(kriging_sf, 32720)

# Guardar resultado
output_path <- file.path(dirname(file_path), "veris_sin_outliers_ec_surface_kriging.geojson")
if (file.exists(output_path)) file.remove(output_path)

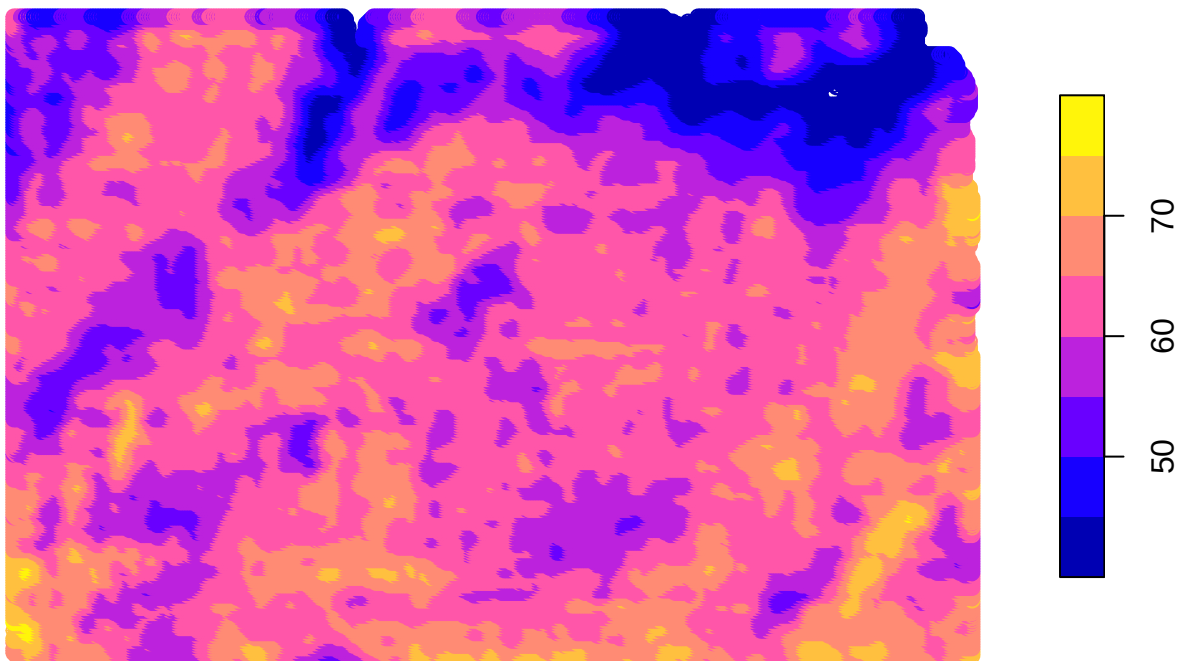
st_write(kriging_sf, output_path, driver = "GeoJSON", quiet = TRUE)
cat("interpolacion completada. Guardado en:\n", output_path, "\n")

# Visualización rapida
plot(kriging_sf["var1.pred"], main = "Interpolacion Kriging - ec_surface")
}

# Ejecutar
kriging_ec_surface("C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh//v

## Procesando: C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh//veris_
## [using ordinary kriging]
## interpolacion completada. Guardado en:
## C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh/veris_sin_outliers.
```

Interpolacion Kriging – ec_surface



Join espacial: Para generar una tabla consolidada de datos, se realiza una unión espacial entre la capa de hexágonos “Grid_con_puntos” y las variables interpoladas provenientes de la capa “veris_data”.

```

setwd("C://Users//Ramiro Manzo//Downloads//pt_rmarckdown//rstudio-project-agro//data_lh")

library(sf)
library(dplyr)

grid <- st_read("soy_performance_completo.geojson")

## Reading layer 'soy_performance_completo' from data source
##   'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\soy_performance_completo'
##   using driver 'GeoJSON'
## Simple feature collection with 10214 features and 22 fields
## Geometry type: POLYGON
## Dimension:      XY
## Bounding box:   xmin: -63.80951 ymin: -33.99375 xmax: -63.7887 ymax: -33.98179
## Geodetic CRS:   WGS 84

veris <- st_read("veris_sin_outliers_ec_surface_kriging.geojson")

## Reading layer 'veris_sin_outliers_ec_surface_kriging' from data source
##   'C:\Users\Ramiro Manzo\Downloads\pt_rmarckdown\rstudio-project-agro\data_lh\veris_sin_outliers_ec_surface_kriging'
##   using driver 'GeoJSON'
## Simple feature collection with 97410 features and 2 fields
## Geometry type: POINT
## Dimension:      XY
## Bounding box:   xmin: 425241.9 ymin: 6238281 xmax: 427146.9 ymax: 6239551
## Projected CRS:  WGS 84 / UTM zone 20S

# CRS
if (st_crs(grid) != st_crs(veris)) {
  veris <- st_transform(veris, st_crs(grid))
}

grid <- grid %>% mutate(hex_id = row_number())

joined <- st_join(veris, grid, join = st_intersects, left = FALSE)

resumen <- joined %>%
  st_set_geometry(NULL) %>% # eliminamos geometría para acelerar agregación
  group_by(hex_id) %>%
  summarise(
    ec_surface = mean(var1.pred, na.rm = TRUE),
    .groups = "drop"
  )

grid_final <- grid %>%
  left_join(resumen, by = "hex_id") # ahora sí, la geometría se mantiene

# st_write(grid_final, "Grid_con_puntos_con_promedios.geojson", driver = "GeoJSON", delete_dsn = TRUE)

grid_final %>%
  as.data.frame() %>% # para que no imprima geometría larga
  select(-geometry) %>%
  head(20)

```

##	Empresa	Campania	Region	Campo	Lote	Actividad	Material
## 1	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 2	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 3	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 4	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 5	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 6	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 7	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 8	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 9	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 10	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 11	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 12	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 13	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 14	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 15	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 16	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 17	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 18	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 19	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
## 20	Garruchos	agropecuaria	2019-2020	Centro	La Magdalena	L4	Soja SojaX
##	Area_lote	Hexagono	Id_Hexagono	Area_hexagono	Lat	Long	
## 1	243.618	1	8cc34d60575ebff	0.02385	-33.98731	-63.80324	
## 2	243.618	2	8cc34d601ce8dff	0.02385	-33.98425	-63.79053	
## 3	243.618	3	8cc34d600af5bff	0.02384	-33.99266	-63.79404	
## 4	243.618	4	8cc34d605393dff	0.02385	-33.98680	-63.80606	
## 5	243.618	5	8cc34d600a76dff	0.02384	-33.99231	-63.79645	
## 6	243.618	6	8cc34d6057b17ff	0.02385	-33.99179	-63.80199	
## 7	243.618	7	8cc34d6050c6bff	0.02385	-33.98920	-63.80752	
## 8	243.618	8	8cc34d60199cdf	0.02385	-33.98420	-63.79759	
## 9	243.618	9	8cc34d6057329ff	0.02385	-33.99128	-63.80480	
## 10	243.618	10	8cc34d6052961ff	0.02385	-33.98568	-63.80343	
## 11	243.618	11	8cc34d60030d3ff	0.02384	-33.99175	-63.79007	
## 12	243.618	12	8cc34d6056b3bff	0.02385	-33.99030	-63.79887	
## 13	243.618	13	8cc34d6054657ff	0.02385	-33.99289	-63.80621	
## 14	243.618	14	8cc34d600b487ff	0.02385	-33.99131	-63.79822	
## 15	243.618	15	8cc34d605634dff	0.02385	-33.98985	-63.80252	
## 16	243.618	16	8cc34d6011b45ff	0.02385	-33.98343	-63.79030	
## 17	243.618	17	8cc34d6051985ff	0.02385	-33.99306	-63.80919	
## 18	243.618	18	8cc34d600ac99ff	0.02384	-33.99055	-63.79272	
## 19	243.618	19	8cc34d6057855ff	0.02385	-33.99115	-63.80201	
## 20	243.618	20	8cc34d600a4abff	0.02385	-33.98981	-63.79510	
##	h3_res12_2019	campaign_year_2019	performance_pct_2019	h3_res12_2021			
## 1	<NA>	2019	18.66481	<NA>			
## 2	<NA>	2019	54.95648	<NA>			
## 3	<NA>	2019	37.04818	<NA>			
## 4	<NA>	2019	38.46908	<NA>			
## 5	<NA>	2019	53.48653	<NA>			
## 6	<NA>	2019	35.54789	<NA>			
## 7	<NA>	2019	40.33041	<NA>			
## 8	<NA>	2019	48.16419	<NA>			
## 9	<NA>	2019	30.26079	<NA>			
## 10	<NA>	2019	41.07490	<NA>			
## 11	<NA>	2019	71.15321	<NA>			

## 12	<NA>	2019	18.96684	<NA>
## 13	<NA>	2019	33.17558	<NA>
## 14	<NA>	2019	46.83522	<NA>
## 15	<NA>	2019	16.75511	<NA>
## 16	<NA>	2019	56.94073	<NA>
## 17	<NA>	2019	20.09122	<NA>
## 18	<NA>	2019	52.63642	<NA>
## 19	<NA>	2019	33.70256	<NA>
## 20	<NA>	2019	57.82286	<NA>
##	campaign_year_2021	performance_pct_2021	h3_res12_2023	campaign_year_2023
## 1	2021	37.44442	<NA>	2023
## 2	2021	54.27125	<NA>	2023
## 3	2021	50.45166	<NA>	2023
## 4	2021	74.70678	<NA>	2023
## 5	2021	62.93598	<NA>	2023
## 6	2021	47.77869	<NA>	2023
## 7	2021	54.18559	<NA>	2023
## 8	2021	54.75749	<NA>	2023
## 9	2021	48.35976	<NA>	2023
## 10	2021	55.25087	<NA>	2023
## 11	2021	83.16163	<NA>	2023
## 12	2021	45.04197	<NA>	2023
## 13	2021	44.11689	<NA>	2023
## 14	2021	63.81562	<NA>	2023
## 15	2021	39.39993	<NA>	2023
## 16	2021	51.53072	<NA>	2023
## 17	2021	36.94627	<NA>	2023
## 18	2021	58.68633	<NA>	2023
## 19	2021	33.75069	<NA>	2023
## 20	2021	50.23164	<NA>	2023
##	performance_pct_2023	hex_id	ec_surface	
## 1	46.84332	1	66.47636	
## 2	73.74146	2	48.23009	
## 3	68.56300	3	64.23605	
## 4	70.93781	4	53.55638	
## 5	73.33036	5	63.60745	
## 6	59.09827	6	68.29024	
## 7	57.32663	7	59.43339	
## 8	68.07064	8	60.31396	
## 9	71.40393	9	62.12487	
## 10	63.37581	10	63.38700	
## 11	78.83390	11	62.38383	
## 12	56.22143	12	60.46754	
## 13	61.70916	13	58.79900	
## 14	73.33338	14	58.18570	
## 15	29.79959	15	65.00516	
## 16	64.53901	16	42.43049	
## 17	50.25227	17	74.00392	
## 18	70.69659	18	64.70842	
## 19	63.71641	19	64.44964	
## 20	80.62896	20	61.27959	